



Company Presentation



Company Presentation



- § Company Overview
- § Production Facilities
- § Quality Control
- § Markets & Products
- § Key Figures
- § New Investments
- § Certifications

- § Forgiatura Morandini was established in 1970 and is a private company belonging to the Morandini family.
- § Located in Civate Camuno (Italy) on a 100,000 sqm area, of which 55,000 sqm are under cover.
- § Continuous focus on a technological and qualitative development of the production of forgings made of special steel qualities and alloys.

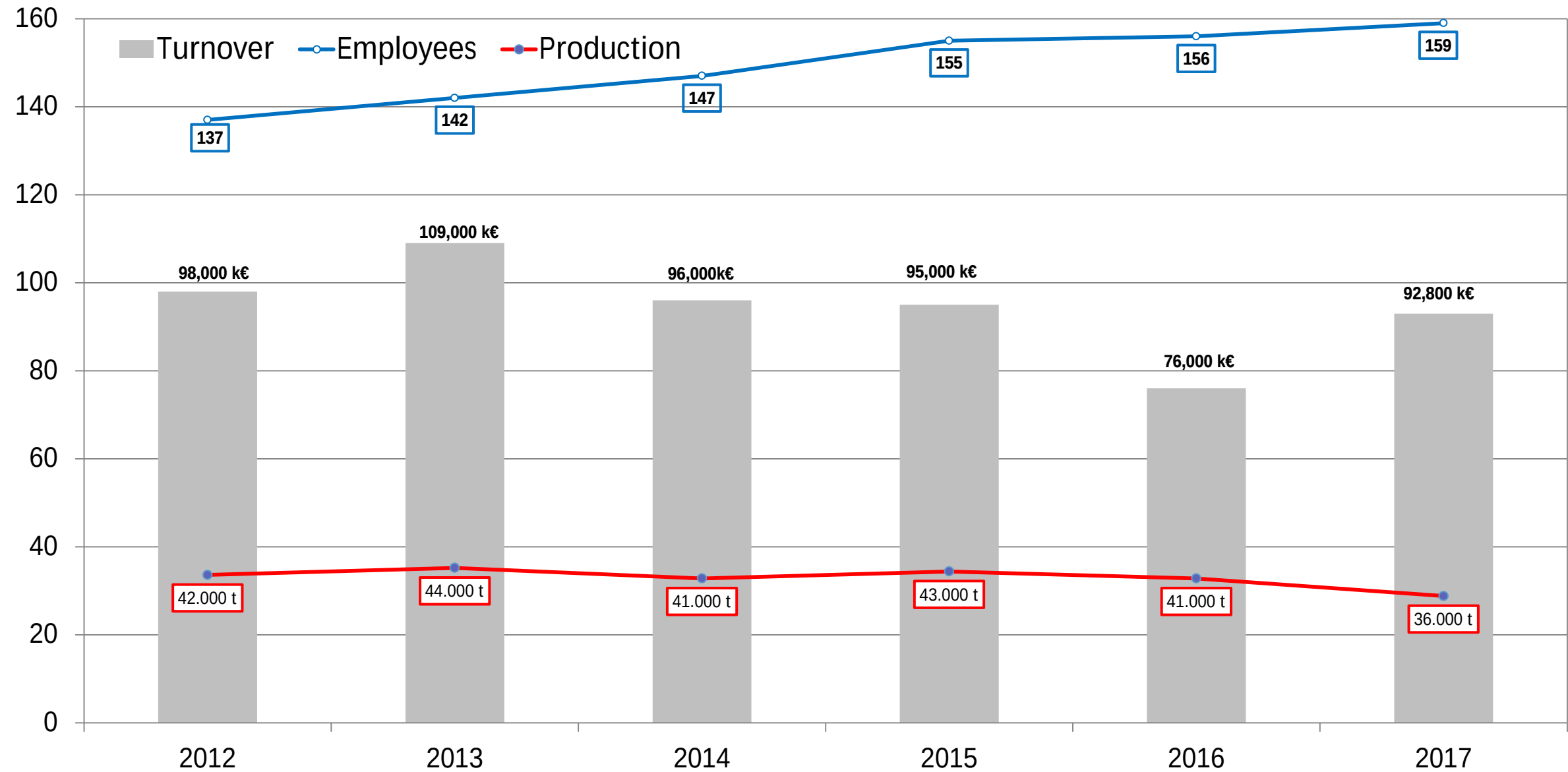


Forgiatura Morandini at a Glance

Forgiatura Morandini

Location	Civate Camuno (Italy)
Production capacity	75,000 t
Markets	Nuclear, Thermal, Hydro, Petrochemical, Mechanical, Naval Industry
Max. ingot size	200 t / 250 t
Melting Processes	Conventional, ESR , VIM , VAR
Forging Presses	16,000 – 7,000 – 3,000 – 3,000 – 1,000 t
Ring mill	Ømax 8,000 mm with Hmax 4,000 mm
Manipulators	300 t - 150 t - 100 t - 80 t - 50 t - 25 t - 20 t - 15 t - 10 t - 3 t
Dimensions	
as forged	Ø 6,500 mm / Lmax = 23,000 mm
as rolled	Ø 8,000 mm / Lmax = 4,000 mm
heat treatment	Ø 8,000 mm / Lmax = 23,000 mm
Heat stability test	✓
Automatic UT	✓
Certificates	ASME III, RINA ISO 9001:2008, UNI EN ISO 14001:2015, BS OHSAS 18001:2007, EnergyMS-59 ISO 50001:2011, LLOYD'S REGISTER OF SHIPPING, AMERICAN BUREAU OF SHIPPING, BUREAU VERITAS, DET NORSKE VERITAS, GERMANISCHER LLOYD, TÜV

Forgiatura Morandini 2012 ÷ 2017



1969 Year of foundation

YEAR

§1970

1978 The first 1,000 ton press

§1980

1988 3,000 ton press

1990 Transfer of property from the founder to his sons

§1990



§2000

2007 7,000 ton press

§2010

2011 Installation of the Ring mill

2012
Alessandro Morandini becomes the sole owner

2016 16,000 ton press

§2020

- § The raw material (blooms and ingots) for the production of the forgings is purchased from qualified steel plants only, which can guarantee vacuum-degassing.
- § Production capability ranges from a minimum of 100 kg to a maximum of 250 t for carbon and alloyed steel, incorporating a minimum diameter of 200 mm, a maximum diameter of 8,000 mm and a maximum length of 23 m.
- § Annual production capacity of 75,000 t



16000 t press

- § 1 oleodynamic 16000 t press
maximum ingot weight: 250 t
- § Railbound manipulator
300 t lifting capacity,
600 t*m loading moment
- § Mobile 150 t manipulator



Production Facilities – Forging Shop

7000 t press

§ 1 oleodynamic 7000 t press

§ maximum ingot weight: 140 t



3000 t press

- § 2 oleodynamic 3000 t presses
- § maximum ingot weight: 35 t



1000 t press

- § 1 oleodynamic 1000 t press
- § maximum ingot weight: 5 t



Production facilities – Ring Mill

Ring Mill

Equipment		status
Ring mill	Ø 8,000 mm H 4,000 mm Max. weight of rolled part: 150 t	operating
Mobile manipulator	150 t	operating
Heating furnaces	Two heating furnaces, load capacity 300 t/each	operating



Heat Treatment Furnaces

- § 7 car-bottom furnaces
- § 5 stationary hearth furnaces
- § 2 overturning furnaces
- § 1 stationary hearth furnace under construction



Quenching

- § 3 Water tanks up to 700 m³
- § 1 Oil Tank
- § 1 Polymer Tank
- § 1 Spray Quenching (horizontal rotation)
- § 2 Forced Air Stations
- § 1 Circular Tank Ø 8 m / 600 m³



Heat Stability Testing Machines

Heat stability 1

- § load capacity: max. 12,000 mm length
- § max. diameter: approx. 2,200 mm
- § max. weight: 150 t

Heat stability 2

- § load capacity: max. 7,000 mm length
- § max. diameter: approx. 1,200 mm
- § max. weight: 35 t
- § Automatic readout



Production Facilities – Machine Shop

- § 22 band sawing machines
- § 2 oxygen lance cutting machines
- § 1 CNC vertical lathe
- § 1 vertical lathe
- § 5 parallel lathes (4 CNC-machines)
- § 1 CNC milling machine (with movable pillar)
- § 1 milling machine (with movable pillar)



Automatic US-Testing System

- § working length: from 1.50 m to 12 m
- § working diameter: from 400 mm to 2,500 mm
- § max. weight: approx. 75 t
- § software UTxx by GE Inspection Technologies



Chemical Analysis

- § optical emission spectrometer ARL 3460
- § portable spectrometer SPECTROPORT F
- § portable X-Ray spectrometers NITON XLT 898
- § Leco instruments for determination of C,S,O,N,H-content

Micrography Inspections

- § optical microscope with digital image manager
- § apparatus for the polishing of test specimens
- § equipment mod. LSP micrography with repetition
- § portable microscope type SM 500

Non-Destructive Testing

- § magnetic particle examination
- § dye penetrant examination
- § manual ultrasonic examination
- § automated ultrasonic examination for shafts
- § visual and dimensional inspection
- § hardness measurers (ERNST/EQUOTIP) HB_{10/3000/15}
- § residual stress determination ASTM E 837
- § radioactivity measurer
- § heat stability testing machine



Mechanical & Technological Properties

- § universal testing machines GALDABINI 40 MN – INSTRON 60 MN (ISO and ASME)
- § impact testing machines up to -196°C
- § testing temperatures up to 600° C
- § drop weight test machine
- § machine for stress rupture testing
- § profile measurer
- § hardness tester
- § portable digital tester
- § portable digital gauger
- § simulated heat treatment of test specimens



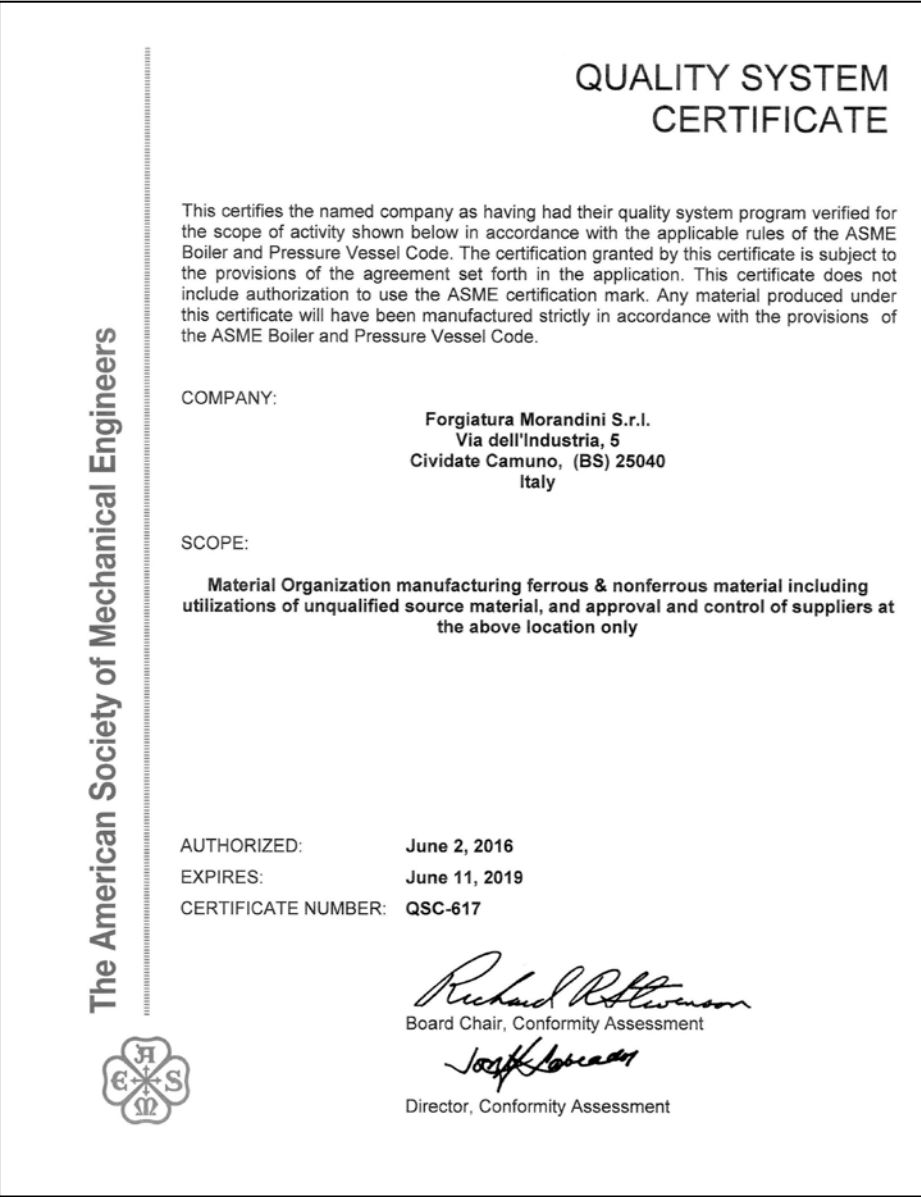
Other Testing Equipment (external)

Forgiatura Morandini strives a strong collaboration with external material testing laboratories for specific tests as

- § SEM (Scanning electron microscope)
- § creep tests
- § fracture mechanics
- § K_{IC} , J_{IC} , CTOD (Crack tip opening displacement)



RINA ISO 9001:2015



ASME III, Div.1 and 3, NCA/WA-3800

Certifications



BS OHSAS 18001 : 2007

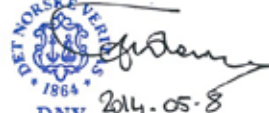


UNI EN ISO 14001 : 2015



EnergyMS-59 ISO 50001 : 2011

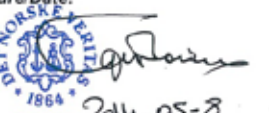
NORSOK Qualification

 FORGIATURA MORANDINI	Qualification Test Record (QTR) NORSOK M-650		QTR02/13		
	Rev. 0	Date 08/05/14			
Manufacturer name Address/web page): Forgiatura Morandini S.r.l. via dell'Industria, 5 - 25040 - Cividate Camuno (BS), Italy www.morandini.it					
Reference standard NORSOK M-650 Ed. 4 September 2011					
Material designation and MDS No Grade and specification: ASTM A182 Gr.F53 MDS D54 Rev.5 – UNS S32750					
Manufacturing summary: MS02-13					
Production and manufacturing process: Open die forgings Production route: MS02-13					
Mandatory conditions and sub-contractors: Heat treated by Forgiatura Morandini S.r.l. according to IT1775/13 and HT02/13.					
Other informations: Mechanical tests shall be performed in laboratories accredited according to ISO/IEC 17025.					
Qualification expires: May 2019					
Tested and Qualified thickness and weight					
Product and Manufacturing Processes:	Test record No.	Tested Thickness mm	Qualified Thickness mm	Test piece Weight kg	Qualified Weight kg
Open die forgings	322	280	Up to 280	320	All
Qualification/Acceptance signature					
Manufacturer:  FORGIATURA MORANDINI Via Dell'Industria, 5 25040 Cividate Camuno (BS)		Prepared by/Date: Bellesi Franco Technical dpt.		Checked by/Date:  Vignola Mario Technical manager	
The manufacturer and this QTR are evaluated and found to be in compliance with the requirements of NORSOK M-650 for supply of the above listed products and materials. This acceptance does not exempt any purchaser from his responsibility to ensure that this qualification is valid for his products within the essential variables of NORSOK M-650.					
Qualified/Accepted by (company name /address): DNV ITALIA SRL Centro Colleoni Palazzo Sirio,2 Viale Colleoni ,9 20864 AGRATE BRIANZA (MB) ITALY			Signature/Date:  DNV MILAN 2014.05.8		

Forgiatura Morandini has been successfully qualified in compliance with the requirements of NORSOK M650 ed. 4 for open-die forgings manufactured in the following steel grades

ASTM A182 F51
thickness up to T=410 mm
NORSOK M630 MDS D44

ASTM A182 F53
thickness up to T=280 mm
NORSOK M630 MDS D54

 FORGIATURA MORANDINI	Qualification Test Record (QTR) NORSOK M-650		QTR01/13		
	Rev. 0	Date 08/05/14			
Manufacturer name Address/web page): Forgiatura Morandini S.r.l. via dell'Industria, 5 - 25040 - Cividate Camuno (BS), Italy www.morandini.it					
Reference standard NORSOK M-650 Edition 4 September 2011					
Material designation and MDS No Grade and specification: ASTM A182 Gr.F51 MDS D44 Rev.5 – UNS S31803					
Manufacturing summary: MS01-13 Rev.3					
Production and manufacturing process: Open die forgings Production route: MS01-13					
Mandatory conditions and sub-contractors: Heat treated by Forgiatura Morandini S.r.l. according to IT1759/13 and HT01/13.					
Other informations: Mechanical tests shall be performed in laboratories accredited according to ISO/IEC 17025.					
Qualification expires: May 2019					
Tested and Qualified thickness and weight					
Product and Manufacturing Processes:	Test record No.	Tested Thickness mm	Qualified Thickness mm	Test piece Weight kg	Qualified Weight kg
Open die forgings	321	410	Up to 410	780	All
Qualification/Acceptance signature					
Manufacturer:  FORGIATURA MORANDINI Via Dell'Industria, 5 25040 Cividate Camuno (BS)		Prepared by/Date: Bellesi Franco Technical dpt.		Checked by/Date:  Vignola Mario Technical manager	
The manufacturer and this QTR are evaluated and found to be in compliance with the requirements of NORSOK M-650 for supply of the above listed products and materials. This acceptance does not exempt any purchaser from his responsibility to ensure that this qualification is valid for his products within the essential variables of NORSOK M-650.					
Qualified/Accepted by (company name /address): DNV ITALIA SRL Centro Colleoni Palazzo Sirio,2 Viale Colleoni ,9 20864 AGRATE BRIANZA (MB) ITALY			Signature/Date:  DNV MILAN 2014.05.8		

Certifications



LLOYD'S REGISTER
OF SHIPPING



RINA



TÜV



TURKISH ATOMIC
ENERGY AUTHORITY



BUREAU VERITAS



DET NORSKE VERITAS
GERMANISCHER LLOYD



AMERICAN BUREAU OF
SHIPPING







§ FORGIATURA MORANDINI srl is
authorised exporter for:

Algeria

Bosnia and Herzegovina

Canada

Israel

Norway

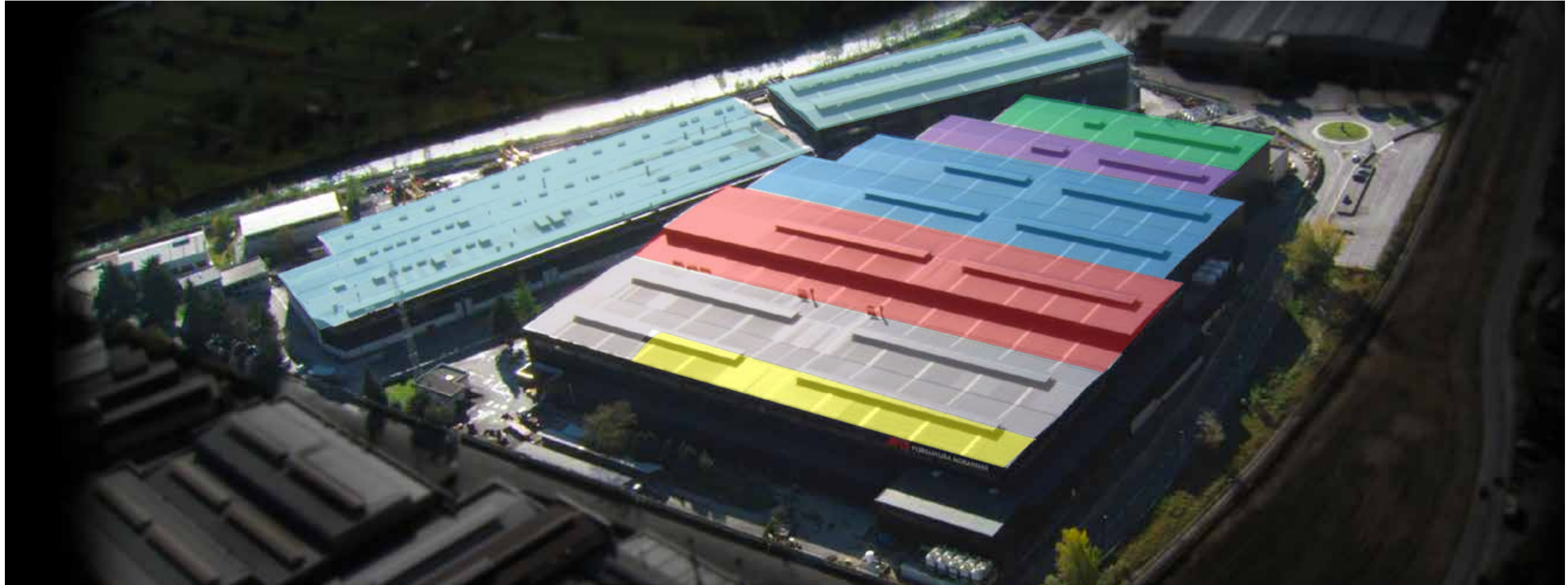
South Africa

South Korea

Switzerland

and Ukraine

Factory Layout

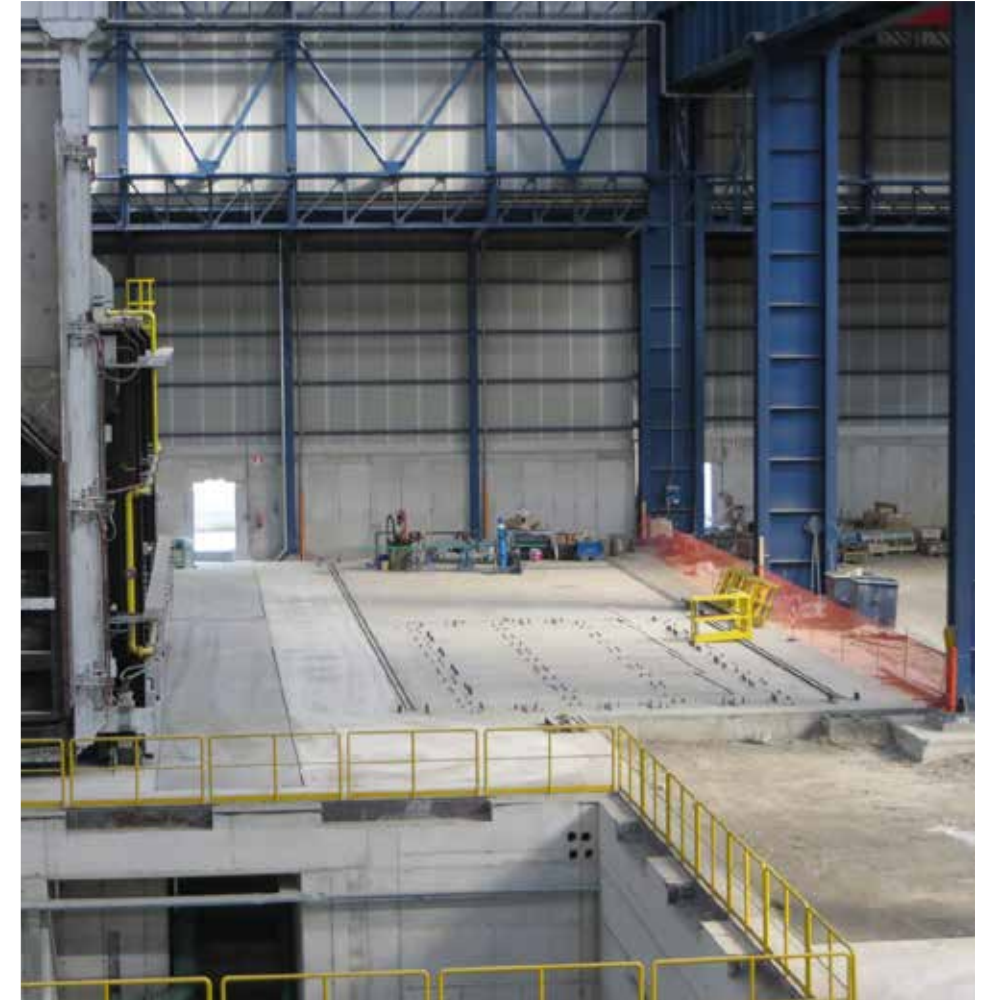


-  FORGING SHOP
-  Laboratories
-  Inspection & shipment

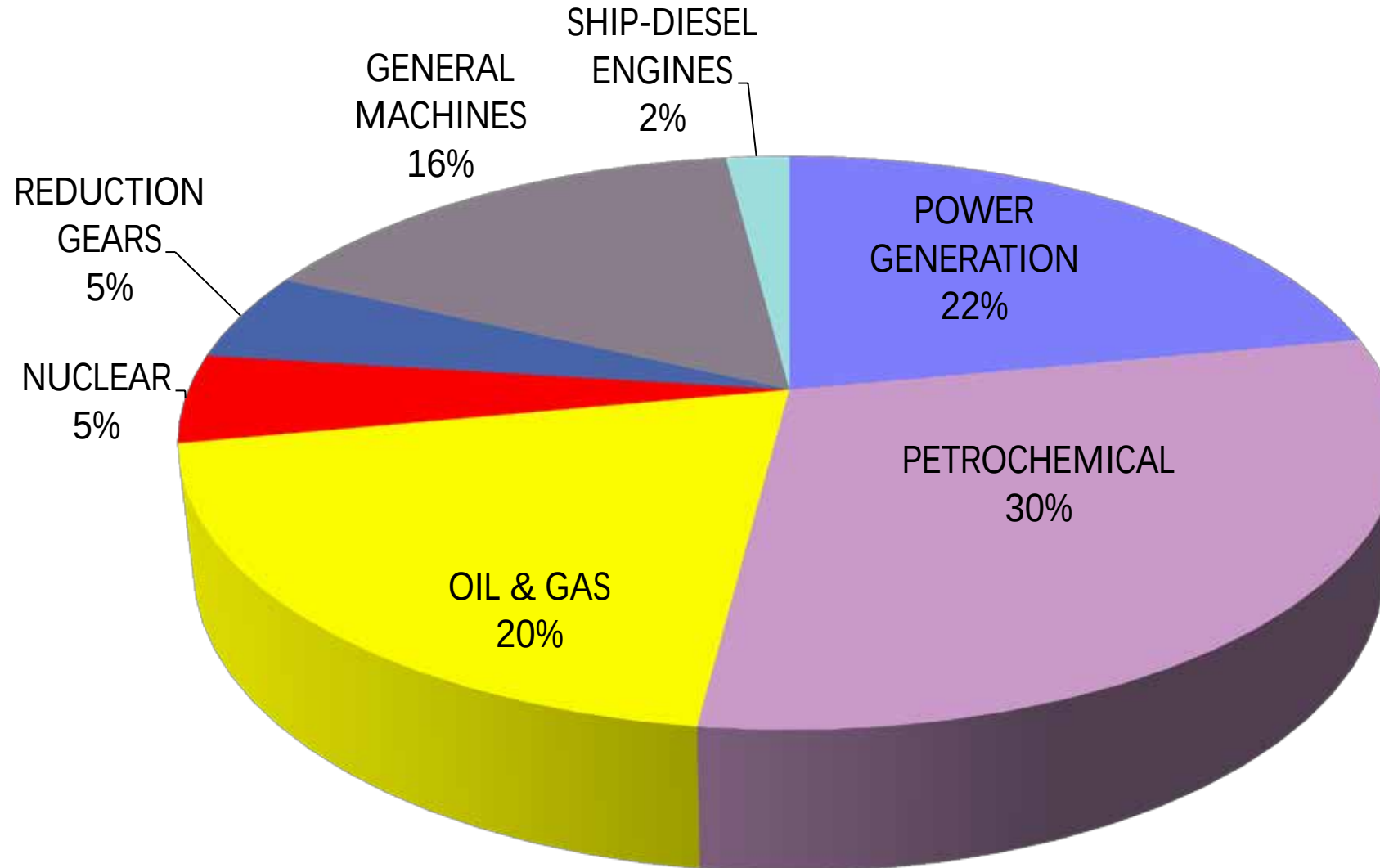
-  RING MILL and 16000 t PRESS
-  Heat treatment – Large rings
-  Cutting – Large pieces
-  Future Investments

Heat Treatment Department

Equipment		Launch
Heat treatment furnace	stationary hearth furnaces: 6.3 x 5.5 x 15 m [W x H x L] 250 t load capacity	Work in progress
Polymer quenching tank	Circular tank (500 m ³) fitted with: 3 dual cooling towers, 6 centrifugal pumps of 30 kW each, temperature control, diameter = 7 m - depth 7 m load capacity from 80 to 100 t	4th quarter 2020



Product mix 2017





**Nuclear Power
Generation**



**Thermal Power
Generation**



**Hydro Power
Generation**



Oil and Gas



**Petrochemical
Equipment**



Compressors & Pumps



Naval Industries



**Mechanical
Industries**

- § Pressurizer
 - § Shell
 - § Upper head
 - § Lower head
- § Steam generator
 - § Shell
 - § Torispherical head
 - § Tube sheets
- § Hot legs
- § Nuclear casks
 - § Shell, Bottom, Cover

Reference Projects:

- § AP1000 - TN International - NPCIL



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

SA508GR.3

forging weight 50 t



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

SA508GR.3

forging weight 44 t



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

10GN2MFA "ASME SA 350 LF5"

forging weight 10 t



SA508GR.3

forging weight 48 t



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

SA508GR.3

forging weight 76 t



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

SA508GR.3

forging weight 45 t



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet

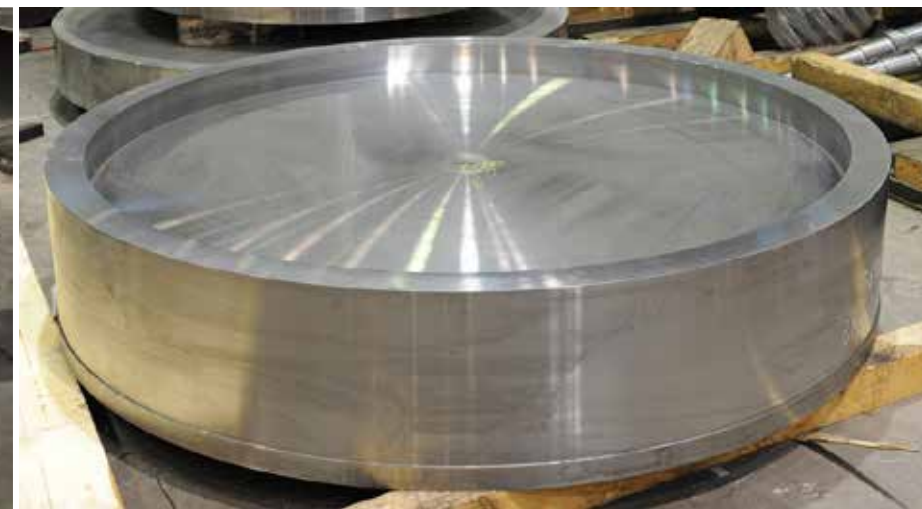
SA508GR.3



SA508GR.3



SA508GR.3



- § Pressurizer - Shell
- § Pressurizer - Upper head
- § Pressurizer - Lower head
- § Steam generator - Shell
- § Steam generator - Torispherical head
- § Steam generator - Tube sheet
- § Nuclear casks
- § Shell, Bottom, Cover

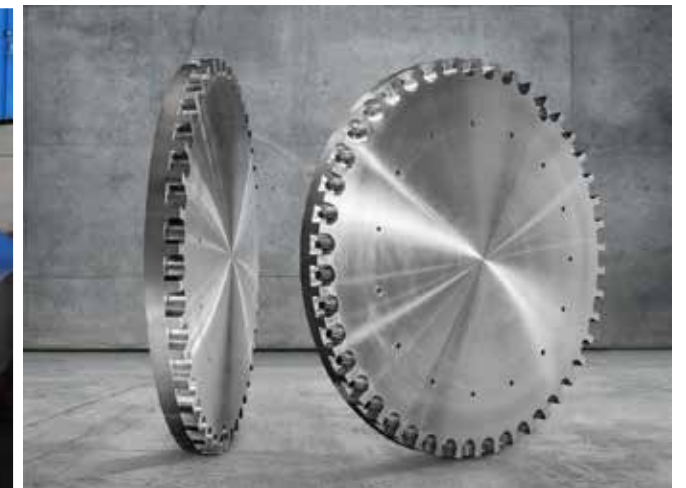
A350LF5



A350LF5

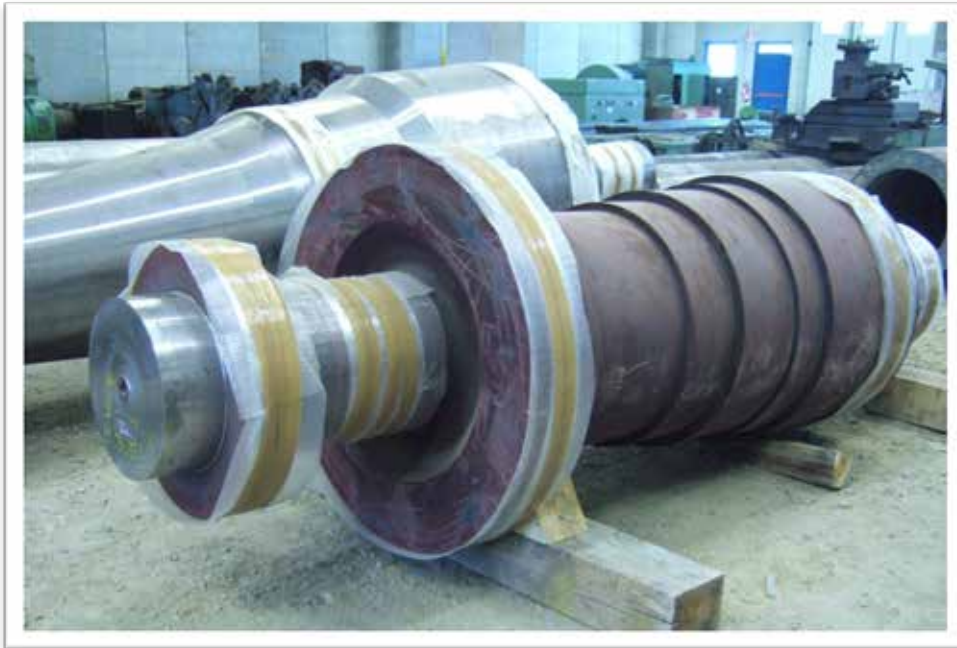


SA182 F316L



- § Gas and steam turbines, either solid or composite
- § Generator shafts





STEAM TURBINE SHAFT

28NiCrMoV12.6.5

Rough machined weight 18 t



STEAM TURBINE SHAFT

28NiCrV8.5

Rough machined weight 15 t



GENERATOR SHAFT

28NiCrMoV13.7.4

Rough machined weight 45 t



GENERATOR ROTOR SHAFT

28NiCrMoV13.7.4

Rough machined weight 40 t



STEAM TURBINE SHAFT

30CrMoNiV4.11

Rough machined weight 50 t



STEAM TURBINE SHAFT

30CrMoNiV4.11

Rough machined weight 33 t



GAS TURBINE END SHAFT

B50 A420 B10

Rough machined weight 2 t



TURBINE SHAFT

30CrMoNiV4.11

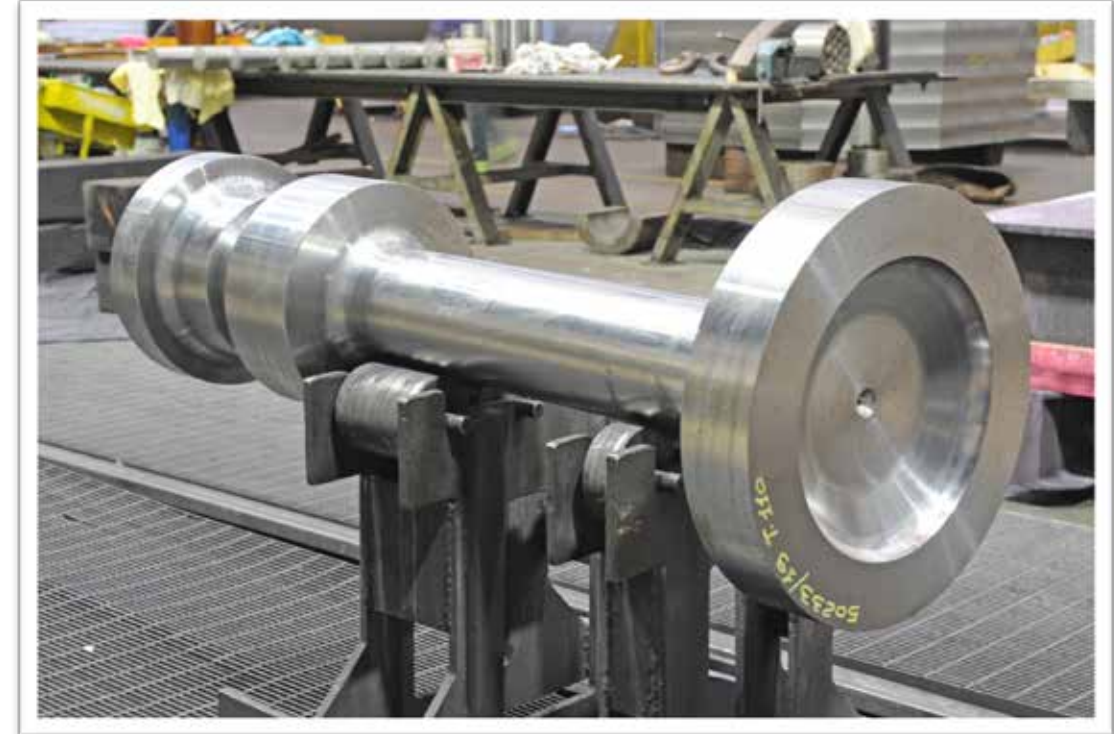
Rough machined weight 3.3 t



GAS ROTOR TURBINE SHAFT

ASTM A471 Type 10

Rough machined weight 1.5 t



GAS ROTOR TURBINE SHAFT

ASTM A471 Type 2

Rough machined weight 1.5 t



GAS TURBINE DISK

B50 A633 D10

Rough machined weight 1 t



GAS TURBINE DISK

B50 A420 B12

Rough machined weight 1.3 t



GAS TURBINE SHAFT

ASTM A471 Type 2

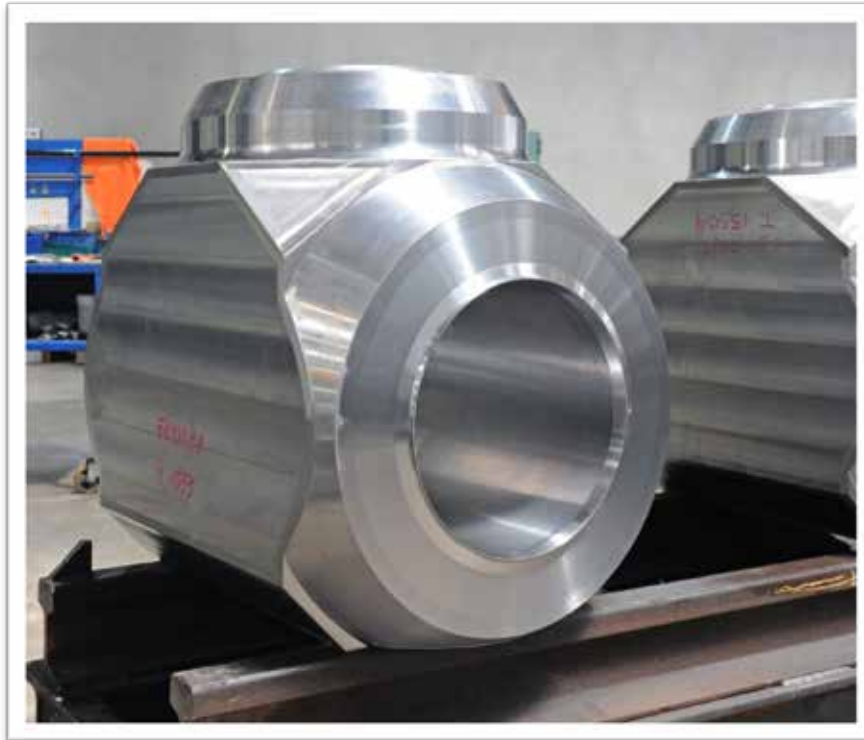
Rough machined weight 2.3 t



TURBINE SHAFT

ASTM A471 Type 10

Rough machined weight 3.5 t



FORGED TEE

A182 F92



FORGED Y-PIECE

A182 F91



PIPES

SA 182 F92

Forged weight 17.2 t
Rough machined weight 10 t

Length 9.5 m



PIPES

ASTM A 335M P91/92/22/11

ASTM A 312M TP347/321

ASTM B564 UNS N08825

- § Generator shafts
- § Hydraulic turbines, generator and coupling shafts
- § Shafts for electrical motors
- § Wind rotors





TURBINE SHAFT

24CrMo5

Rough machined weight 54 t



TURBINE SHAFT

24CrMo5

Rough machined weight 54 t



TURBINE SHAFT

34CrNiMo6

Forging weight 92 t



TURBINE SHAFT

34CrNiMo6

Rough machined weight 64 t

Forging of shaft



Final inspection of shaft



TURBINE SHAFT

A668 Mcl.D

Forging weight 68 t



LOWER SHAFT

ST442S-25CrMo4

Forging weight 75 t



BULB TURBINE SHAFT

A668 cl.E

Rough machined weight 76 t



NDE SHAFT

A668 cl.E

Rough machined weight 47 t



ROTOR BODY

25CrMo4

Rough machined weight 30 t



SHAFT SR

42CrMo4

Rough machined weight 22 t



TURBINE SHAFT

P355QH1

Forging weight 95 t



TURBINE SHAFT

23CrNiMo7.4.7

Fully machined weight 14 t



THRUST COLLAR

A668M cl. E

Forging weight 60 t

- § Valve bodies
- § Pump bodies
- § Spheres
- § Covers
- § Flanges
- § Tubes





24" Grip Lock Connector

A105

Rough machined weight 10.3 t



30" Quali Tee Tap Connector

A694 F65

Rough machined weight 7.8 t



RISER

2¼ Cr 1 Mo

Forging weight 30 t



RISER

2¼ Cr 1 Mo

Forging weight 32 t



B.O.P

2¼ Cr 1 Mo

Rough machined weight 16.5 t



B.O.P

2¼ Cr 1 Mo

Rough machined weight 12.3 t



BLOCK MASTER VALVE

2¼ Cr 1 Mo

Forged weight 23.6 t



BLOCK MASTER VALVE

2¼ Cr 1 Mo

Rough machined weight 13.5 t



PINION SHAFT

AISI 4330 mod

Rough machined weight 3.1 t



VALVE BODY

2¼ Cr 1 Mo

Rough machined weight 2.7 t



STRESS JOINT

2¼ Cr 1 Mo

Forging weight 20 t



STRESS JOINT

SA 336 2¼ Cr 1 Mo

Rough machined weight 13.4 t

Length 12.8 m



LOWER BODY

AISI 8630

Rough machined weight 4.6 t



UPPER BODY

AISI 8630

Rough machined weight 3.1 t



JACKET

ASTM A 182M F22 CL.3

Rough machined weight 2 t



LIFT CAP

2¼ Cr 1 Mo

Rough machined weight 1.8 t

- § Tube sheets
- § Channel shells
- § Covers
- § Bottoms
- § Bushes
- § Flanges
- § Nozzles
- § Valves
- § Stub-ends
- § Pinions
- § Tubes





SHELL

SA336 F22V

Forged weight 68 t

Final dimensions: Ø5027/4679x2237mm





SHELL

SA336 F22V

fully machined 81 t



SHELL

SA336 F11 Cl.3

fully machined 53 t



NOZZLE 68"

SA508 Gr.3 Cl.1

Forging weight 10 t



INLET-FLANGE

SA508 Gr.3

Rough machined weight 12.5 t



SEPARATOR

ASME SA336M F347

Machined weight 17 t



LATERAL NOZZLE

ASTM A182 F22

Machined weight 9.5 t



FORGING SHELL

12CrMo9.10

Rough machined weight 40 t



SPECIAL LIFTING FLANGE

A508 Gr3 Cl.2

Rough machined weight 7 t



HEAT EXCHANGER

SA 336 F22 CL3

Rough machined weight 10.3 t



HEAT EXCHANGER

SA 336 F22 CL3

Rough machined weight 15 t



CASING

15NiCrMoVb5

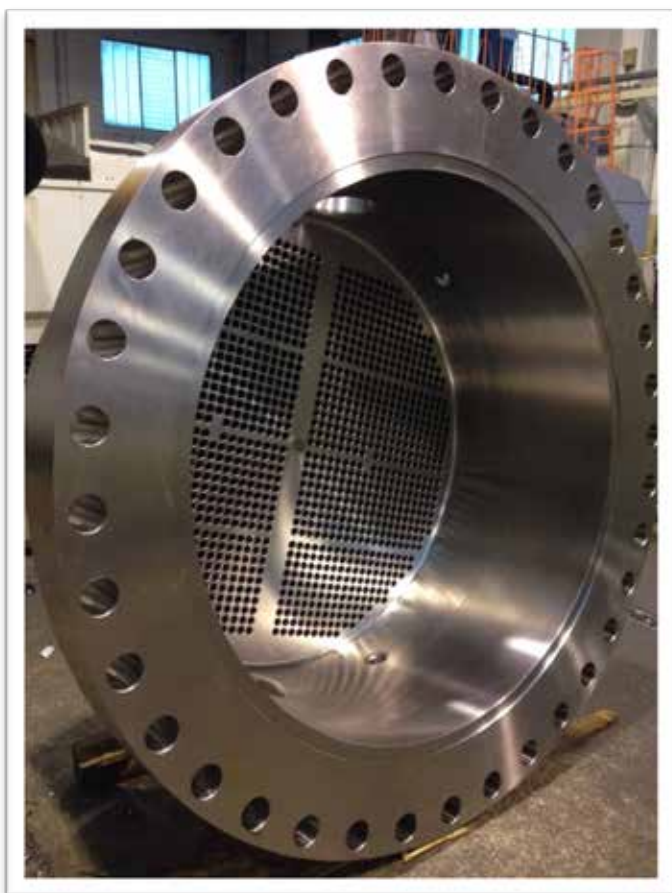
Machined weight 11.3 t



SHELL

SA336 F11 cl.2

Forging weight 28 t



TUBESHEET

SA182 F60

Machined weight 5.8 t



TUBESHEET

SA182 F60

Machined weight 4.6 t



TUBESHEET

SA965 F304H

Machined weight 15.9 t

- § Channel shells
- § Covers
- § Bottoms
- § Bushes
- § Flanges
- § Nozzles
- § Valves
- § Stub-ends
- § Pipes





VALVE BODY

X3CrNiMo13-4

Fully machined weight 5.5 t



CONNECTING ROD

42CrMo4

Rough machined weight 2 t



SPHERICAL BODY

SA350 LF2

Machined weight 12 t



COMPRESSOR HOUSING

SA350 LF6

Forging weight 45 t



CASING

A350 LF2

Machined weight 14,5 t



NOZZLE

A350 LF3

forging weight 9 t



SHELL

ASME SA 336M F22 CL.3

Rough machined weight 20 t



HIGH-PRESSURE COMPRESSOR

42CrMo4

Rough machined weight 12.5 t



PUMP BODY

A182 F53

Welded & fully machined 10 t

- § Propeller shafts
- § Intermediate shafts
- § Thrust-bearings and rudder shafts
- § Components of marine diesel engines such as connecting rods, piston rods, piston heads, cylinder covers, cross heads





CONNECTING RODS

C25

Machined weight 4 t



PROPELLER SHAFT

CK45

Machined weight 28 t



INTERMEDIATE SHAFT

30Mn5

Forging weight 65 t



INTERMEDIATE SHAFT

30Mn5

Rough machined weight 51 t



RUDDER TRUNK

Carbon steel

Machined weight 40 t



PROPELLER SHAFT

12NiCrMo7

Machined weight 19 t



PROPELLER SHAFT

28Mn6

Forging weight 60 t



PROPELLER SHAFT

C45

Machined weight 22 t

- § Centrifugal casting moulds
- § Pinions
- § Gears wheels
- § Shafts
- § Columns for presses
- § Cylinder covers and piston heads
- § Die blocks
- § Components for extrusion equipment such as casings, cylinder liners, headers, matrixes





MOULD

C25

machined weight 25 t



FLANGE SHAFT

42CrMo4

machined weight 11.5 t



CRANK SHAFT

42CRMo4

machined weight 10 t



ROLL

42CrMo4

forging weight 61 t



CYLINDER

21NiCrMo3

Rough machined weight 64.5 t



FLANGE YOKE

34CrNiMo6

Rough machined weight 4 t



GEAR PINION SHAFT

40NiCrMo7

machined weight 15 t



CONTAINER

Container Mantle : X38CrMoV5-1 (1.2343)

Intermediate Liner : X38CrMoV5-3 (1.2367)

Inner Liner : X38CrMoV5-1 (1.2343)

Total weight after assembling 68 t



BOTTOM COVER PLATE

SA266 CL.2

machined weight 0.2 t



FREMM

S355 J2G3

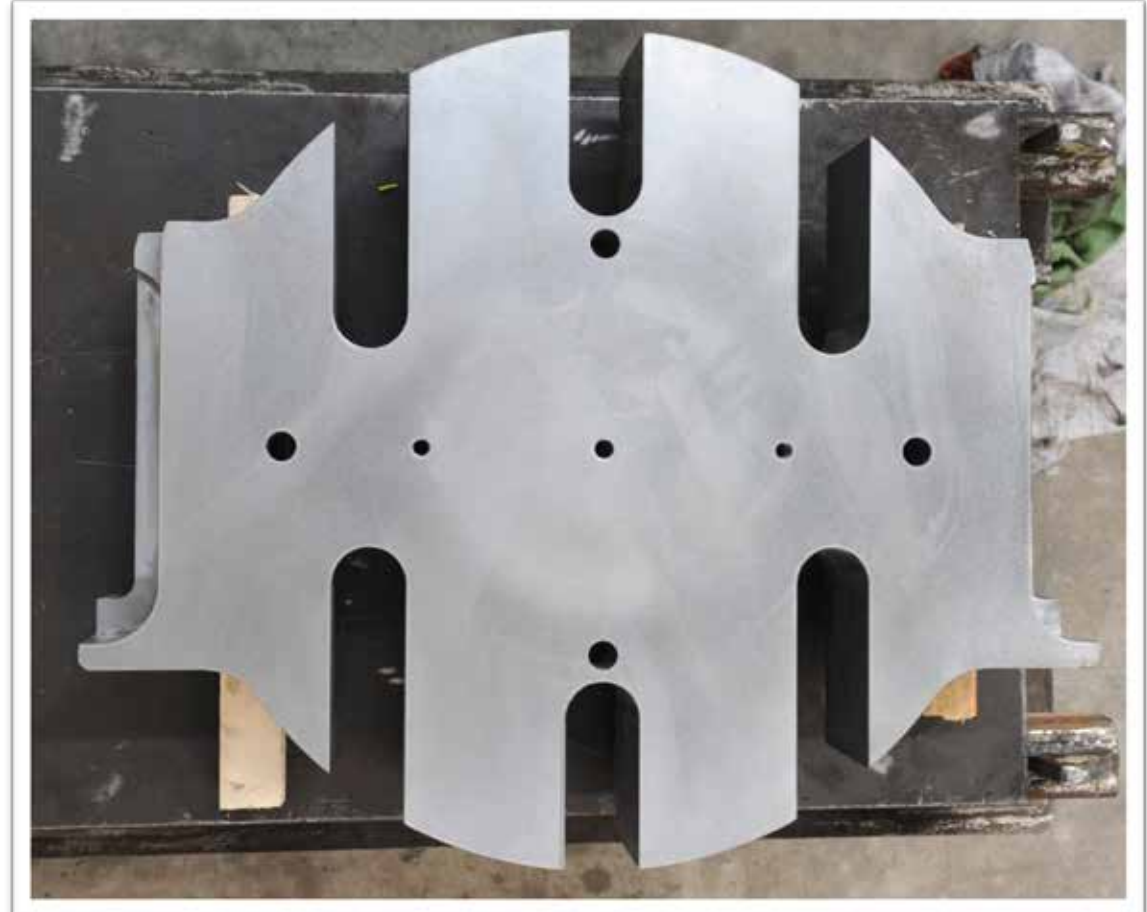
machined weight 3.5 t



CYLINDER CROSSHEAD

S355 J2G3

machined weight 35 t



TOP COVER EXTENSION PLATE

SA266 CL.2

machined weight 0.2 t



ANVIL RING

34CrNiMo5

Rough machined weight 126 t



Thanks for your attention!

For further information please visit our homepage
or contact us directly.



Web site: www.morandini.it

Contact: info@morandini.it

Phone: +39 0364 347 0

